Exercice 3.2 Global

```
func _init():
    if OS.get_name() == "Windows":
        print("OK")
    else:
        print("Not supported")
```

Exercice 3.5 Signals and random

Enemy.gd

```
@export var color: Color = Color(1,1,1,1): get = _get_color, set =
_set_color

func _ready():
    $AnimatedSprite2D.modulate = color
    move_along_path()

func _set_color(value):
    color = value

func _get_color():
    return color
```

TileGenerator.gd

```
func spawn_new_enemy(position):
    if roll_dice(0.9):
        emit_signal("spawn_enemy", position, Color(1, 1, randf_range(0,
1)))
```

Game.gd

```
func on_spawn_enemy(position, color):
    var enemy = Enemy.instantiate()
    enemy.color = color
```

Exercice 3.6 Tile Generation - Add a New Tile Type

TileGenerator.gd

```
enum TileType { Empty = -1, Ground = 0, Hole = 1 }

func generate_next_tile_type(_pos, prevType):
    if prevType != TileType.Hole and prevType != TileType.Empty and
roll_dice(0.6):
    return choose(0.7, TileType.Hole, TileType.Empty)
    else:
        return TileType.Ground
```

Game.gd

```
func check_player_position_on_map(position):
    var coords = tileMap.get_cell_atlas_coords(0, position)
    var tile = self.atlas_coords_to_tile_type(coords)
    if tile == tiles_gen.TileType.Empty or tile ==
tiles_gen.TileType.Hole:
        game_over()

func tile_type_to_atlas_coords(tileType: int):
    match tileType:
        tiles_gen.TileType.Empty:
            return Vector2i(-1, -1)
        tiles_gen.TileType.Ground:
            return Vector2i(0, 0)
        tiles_gen.TileType.Hole:
            return Vector2i(3, 0)

func atlas_coords_to_tile_type(coords: Vector2i):
    match coords:
        Vector2i(-1, -1):
            return tiles_gen.TileType.Empty
        Vector2i(0, 0):
            return tiles_gen.TileType.Ground
        Vector2i(3, 0):
            return tiles_gen.TileType.Hole
```

Exercice 3.7 - Collision Detection and Invincibility

Game.gd

```
func _on_collided_with_enemy(_other: Node):
    if not moving and not isInvicible:
        game_over()

func _on_moving_along_path_finished(destination):
    moving = false
    set_player_look_at(tiles_gen.front())
    check_player_position_on_map(destination)
```

```
    isInvicible = true
    await get_tree().create_timer(0.5).timeout
    isInvicible = false
```

Exercice 3.8 Tweens

In ScoreLabel.gd

```
func update_score():
    text = String.num(global.score)
    bounce()

func bounce():
    var tween = create_tween()
    tween.tween_property(self, "scale", Vector2(1.3, 1.3), 0.1)
    tween.tween_property(self, "scale", Vector2(1.0, 1.0), 0.15)
```